Errata

On Coppock and McClellan (2019), *Research and Politics* 6(1)

This note lists nine corrections to "Validating the demographic, political, psychological, and experimental results obtained from a new source of online survey respondents," published in *Research and Politics* 6(1), doi 10.1177/2053168018822174. Two are counts of statistically significant differences in the online appendix, one reverses the direction of a descriptive comparison, one is a count off by one, two are sample sizes in the appendix study manifest, one is the bootstrap behind appendix Table 1, which the deposited code leaves unseeded and runs too few times to settle the third decimal it prints, one is a missing space in the appendix study manifest, and one supplies a word the Discussion drops. It also carries three entries of a second kind, which quote a passage and propose no change to it: the p-value column of appendix Table 1, which is not the two-tailed test the appendix describes, one pair of sentences on the same page that disagree about whether Lucid or MTurk is closer to the ANES on voter turnout, and one pair of sentences that name a 2016 General Social Survey the deposit does not hold. **No correction and no note here changes a conclusion of the paper.** The paper's finding, that Lucid respondents track US national benchmarks well and can serve as a drop-in replacement for Mechanical Turk, is unaffected, as is its one negative result on the healthcare rumor corrections.

Every number in a corrected sentence below is computed from the deposited replication archive (10.7910/DVN/DDWWJW) at the time this document is rendered, including the numbers that were not wrong. Bold marks the corrections. The code and output supporting each correction are at github.com/active-maintenance-aec/coppock_mcclellan_2019. Entries 8 and 9 are the exception to all three: neither states a number, the evidence for both is the published PDF rather than the archive, and both were found by setting this edition's pages beside the published ones rather than by running anything.

The first two entries share a cause, and the seventh is why they no longer rest on it. Section 1 of the online appendix states that "we obtain two-tailed p-values under a normal approximation," and the deposited `12_DistanceTests.R` instead computes `pnorm(2 * (1 - abs(z)))`, which is not a p-value for any hypothesis. The two expressions differ in where they put the threshold: the deposit's calls a difference significant at the 0.05 level once `|z|` exceeds 1.82, where the two-tailed test needs 1.96. Read off the standard errors appendix Table 1 prints, extraversion is the only one of the 21 characteristics whose verdict falls between them, and the corrections below are what follows for the two sentences that count significant differences. Those printed standard errors are themselves a single draw from an

unseeded bootstrap, which is entry 7. Under the seeded 125,000-resample bootstrap this note now runs, extraversion clears neither threshold, so both corrections hold whichever expression is used, and both are also recomputed from the distance and the standard error the published table prints, which do not move. The p-value column that the cause produced is itself entry 10, which quotes the sentence that describes it and corrects nothing.

1. Lucid is not significantly closer than MTurk on all five personality traits

> Formal hypothesis tests demonstrate that Lucid is significantly closer to the ANES 2012 than MTurk on **all five traits**.

Corrected:

> Formal hypothesis tests demonstrate that Lucid is significantly closer to the ANES 2012 than MTurk on **four of the five traits, extraversion excepted**.

Appendix Table 1 prints 0.100 and 0.052 for extraversion's distance and bootstrapped standard error. Those two numbers give $\mathtt{z} = 1.92$ and a two-tailed p-value of 0.054, above the 0.05 threshold; the table's own p-value column prints 0.034, which is the deposit's expression rather than the two-tailed test. The maintained rewrite's seeded run reaches the same conclusion, at $\mathtt{z} = 1.79$ and $\mathrm{p} = 0.074$. The other four traits clear 0.05 by a wide margin under either test.

2. The number of significant differences in the online appendix

> Out of 21 opportunities, Lucid is closer to the ANES 18 times, **14** of which are significant.

Corrected:

> Out of 21 opportunities, Lucid is closer to the ANES 18 times, **13** of which are significant.

Extraversion is the difference between the two counts, for the reason given above. Recomputing every p-value from the distance and standard error appendix Table 1 prints gives the same 13. The sentence that follows it is unaffected: MTurk is closer in 3 instances and significant in one, voter turnout, under both tests.

3. Six demographic differences are significant, not five

> Out of the 11 demographic variables that are measured for both Lucid and MTurk, the Lucid mean is closer to the ANES mean in nine instances, **five** of which are statistically significant.

Corrected:

> Out of the 11 demographic variables that are measured for both Lucid and MTurk, the Lucid mean is closer to the ANES mean in nine instances, **six** of which are statistically significant.

Appendix Table 1 prints a p-value below 0.05 for Female, Education, Age, Mean income, Black and South, and Lucid is the closer sample on all six. The count is the same under the two-tailed test corrected in the first two entries, so it does not turn on which p-value is used.

4. The sample size of the Lucid welfare replication

> Sample: **1,811** Lucid respondents.

Corrected:

> Sample: **3,294** Lucid respondents.

The sentence is in section 2.1.3 of the online appendix. 1,811 is the Lucid sample of the Hiscox free-trade replication, reported in section 2.4.2 of the same appendix. The welfare data hold 3,504 Lucid rows, of which 3,294 have a usable outcome, and the manifest reports the analysed sample in every other section.

5. The sample size of the original Kam and Simas study

> Sample: **761** respondents gathered from a Knowledge Networks national probability sample.

Corrected:

> Sample: **752** respondents gathered from a Knowledge Networks national probability sample.

The sentence is in section 2.3.1 of the online appendix. Appendix Table 2 reports N = 752 for all three models fitted to this sample, and 752 is what the deposited data give. The manifest reports the analysed sample rather than the source study's throughout: its entry for the 1984 General Social Survey gives 943, where the deposited file holds 1,473 respondents for that survey.

6. The MTurk sample under-represents southerners

> The regional balance on Lucid comes very close to the 2012 ANES, whereas the MTurk sample appears to **overrepresent** southerners.

Corrected:

> The regional balance on Lucid comes very close to the 2012 ANES, whereas the MTurk sample appears to **underrepresent** southerners.

The southern share is 30.1 per cent on MTurk against the ANES 2012 benchmark of 37.2, so MTurk sits 7.1 points below it; Lucid is at 37.6. Figure 1, which is where the sentence sends the reader, plots the same thing in standard deviations: MTurk at –0.146 and Lucid at 0.009, with the benchmark at zero. MTurk over-represents the Northeast and the Midwest instead. Appendix Table 1 is unaffected: it reports the distance between the two samples rather than the direction of either, and its South row is right.

7. The standard errors and p-values of appendix Table 1 are one draw from an unseeded bootstrap

> Table 1's Bootstrapped SE and p-value columns are **one draw from a 100-resample bootstrap for which the deposited code sets no seed**. Neither column is reproducible: running the deposited procedure again moves it, and moves the count of significant differences the appendix states along with it.

Corrected:

> **This edition prints a 125,000-resample bootstrap at a declared seed**. All 21 distances are unchanged and reproduce the published column exactly; thirteen of the 21 standard errors and ten of the 21 p-values differ from it at the three decimals the table prints.

The sampling error of a bootstrap standard deviation is about `SE / sqrt(2B)`. At the deposit's `B` = 100 that is roughly 0.0035 on a standard error near 0.05, three times the last digit the table prints, which is why `appendix_table_1_bootstrap_stability.R` recovers the published count of 14 significant differences on only 39.5 per cent of 200 seeds, ranging 12 to 15. At `B` = 125,000 it falls to 0.00015, about a seventh of that last digit, and the printed column stops being a property of the seed. Nothing here moves a distance, a sign, or the size of any difference, which is what the appendix's argument rests on. What it does settle is the two counts corrected in entries 1 and 2: at this precision they no longer turn on which of the two p-value expressions is used, because extraversion falls short of 0.05 under both.

8. A missing space in the healthcare rumor study manifest

> **Control:(No** introduction before asking the rumor belief question.)

Corrected:

> **Control: (No** introduction before asking the rumor belief question.)

The item is in section 2.5.1 of the online appendix. The published page sets no space between the bold `Control:` and the parenthesis after it, where the parallel item in section 2.4.1 sets one and so does every other labelled condition in the manifest. It is the page's own spacing and not an extraction artifact: read off the rendered PDF at 300 dpi, the bold run ends and the roman run begins at the same horizontal position, against a gap of one space width in section 2.4.1. No comparison of text can see it, because a space is not a token.

9. A missing word in the Discussion

> While we think the theories that underlie most survey experiments conducted in the USA would extend to Lucid, we emphatically do not mean **to** that any experiment conducted on a convenience sample is relevant for theory.

Corrected:

> While we think the theories that underlie most survey experiments conducted in the USA would extend to Lucid, we emphatically do not mean **to imply** that any experiment conducted on a convenience sample is relevant for theory.

The published page sets "do not mean to that", which is ungrammatical: a verb is missing after "to". The authors' own LaTeX source for the article, in the archived working folder, reads the same, so the omission is not the publisher's typesetting. The paragraph around it separates a theory's predictions extending to a sample from that sample being drawn at random from a population, and the sentence is the authors declining the stronger reading of their own argument, so the missing verb is one of "imply", "say" or "suggest". "Imply" is the one supplied here and the choice is recorded rather than presented as a reading of the page: nothing in the paragraph turns on which of the three it is.

Carried, and not corrected

The three entries below are different in kind from the nine corrections above. Each quotes a passage this edition reprints exactly as the paper published it, and proposes no change to it.

In each, no corrected token can be set without writing the authors' choices for them. The turnout sentences cannot both stand, and each is supported by a different deposited file, so correcting either would mean choosing between two ANES benchmarks. The year in

the heterogeneity section is wrong, but the deposited file that supplies the panel does not say which of two surveys it holds, so a replacement year could not be verified. The p-value column of appendix Table 1 is wrong against the test the appendix describes, but its twenty-one cells are the same cells entry 7 already changes, and a second correction to them would set two values in one cell.

Each is recorded here because an edition that silently reproduced them would leave a reader unable to tell a faithful reprint from an uncorrected error.

10. The p-value column of appendix Table 1 is not the two-tailed test the appendix describes

> Because no formula for the standard error of this procedure is available, we use the nonparametric bootstrap to estimate sampling variability and we obtain **two-tailed p-values under a normal approximation**.

The sentence sets the method on the same page as the table whose last column it describes, and the column is not that test. The deposited `12_DistanceTests.R` computes `pnorm(2 * (1 - abs(z)))`, so the published column is that expression read at the distance and standard error the table prints, and the preamble gives the cause and the threshold it moves. Against the two-tailed p-value computed from those same two printed numbers, the column's 21 cells read, in the table's order, Female 0.000 against 0.001; Education 0.000 against 0.000; Age 0.000 against 0.000; Mean income 0.000 against 0.003; White 0.873 against 0.661; Black 0.000 against 0.000; Hispanic 0.357 against 0.230; Northeast 0.179 against 0.143; Midwest 0.409 against 0.258; South 0.010 against 0.032; West 0.948 against 0.858; Voter registration 0.157 against 0.129; Voter turnout 0.000 against 0.005; Party ID 0.010 against 0.033; Ideology 0.000 against 0.000; Political Interest 0.000 against 0.000; Extraverted 0.034 against 0.054; Agreeable 0.001 against 0.008; Conscientious 0.000 against 0.000; Stable 0.000 against 0.000; Open 0.005 against 0.023. The two columns disagree about significance at 0.05 for one characteristic, extraversion, which entry 1 corrects in the sentence that counts it, and entry 2 corrects the count that follows from it. No corrected column is set here. Entry 7 already changes how the bootstrap behind the same cells is drawn, so a second correction to them would put two values in one cell, and which statistic the table should report is a choice for the authors and not a token the page supplies. No conclusion of the paper changes: Lucid is closer to the ANES on eighteen of the 21 characteristics under either column, the large differences sit where the paper puts them, and extraversion is the one characteristic the choice of expression moves across the threshold.

11. Two sentences on one page disagree about which sample is closer on voter turnout

> Voter registration and turnout seem to vary somewhat across samples, with **the Lucid sample corresponding more closely to the 2012 ANES baseline for voter registration and voter turnout**.

> Lucid is significantly closer to the ANES 2012 on party identification, ideology, and political interest, while **MTurk is significantly closer for voter turnout**.

The two sentences are on one page and disagree about turnout. Each is supported by a file the authors deposited, and the two files carry different ANES 2012 turnout benchmarks. The standardized file the distance tests read puts it at 0.702, which makes MTurk the closer sample and gives appendix Table 1 its distance of -0.124, a two-tailed p-value of 0.005 from the page's own distance and standard error. The political table's weighted mean puts the same benchmark at 0.756, against Lucid's 0.763 and MTurk's 0.706, which makes Lucid the closer sample. Voter registration is unaffected. Correcting either sentence would mean choosing between two deposited benchmarks, which is an analytical decision about what the ANES baseline is and not a token the page supplies, so the edition reprints both as published. No conclusion of the paper changes: Lucid is closer on party identification, ideology, and political interest under either benchmark, and the registration comparison favours Lucid under both.

12. The welfare heterogeneity section names a 2016 GSS the deposit does not hold

> We assess whether the treatment effect of receiving the "assistance to the poor" versus "welfare" phrasing varies among white, black, and Latino respondents, among both the Lucid sample and respondents in the **2016 GSS**.
>
> While the point estimates for the **2016 GSS** CATEs are nearly identical, in the Lucid sample white respondents exhibited larger treatment effect magnitude than black or Latino respondents, though again these differences are not statistically significant.

The deposited welfare data hold the 1984 and 2014 General Social Surveys and no other, 943 and 2,457 usable respondents, and the article's own Experiment 1 section names the 2014 GSS as the baseline. The deposited `standardizedExperimentsHeterogeneousEffects.RData`, which supplies the original panel of Figure 3, is a pre-computed table whose comparison sample is labelled "Original" with no year and which no deposited script builds, so which of the two surveys the panel draws on cannot be established from the deposit. A corrected sentence would have to name a year this edition cannot verify, so both sentences are reprinted as published. No conclusion of the paper changes: the sentences report low heterogeneity among white, black, and Latino respondents in both samples, a statement about the CATEs and not about the year of the survey that supplies them.